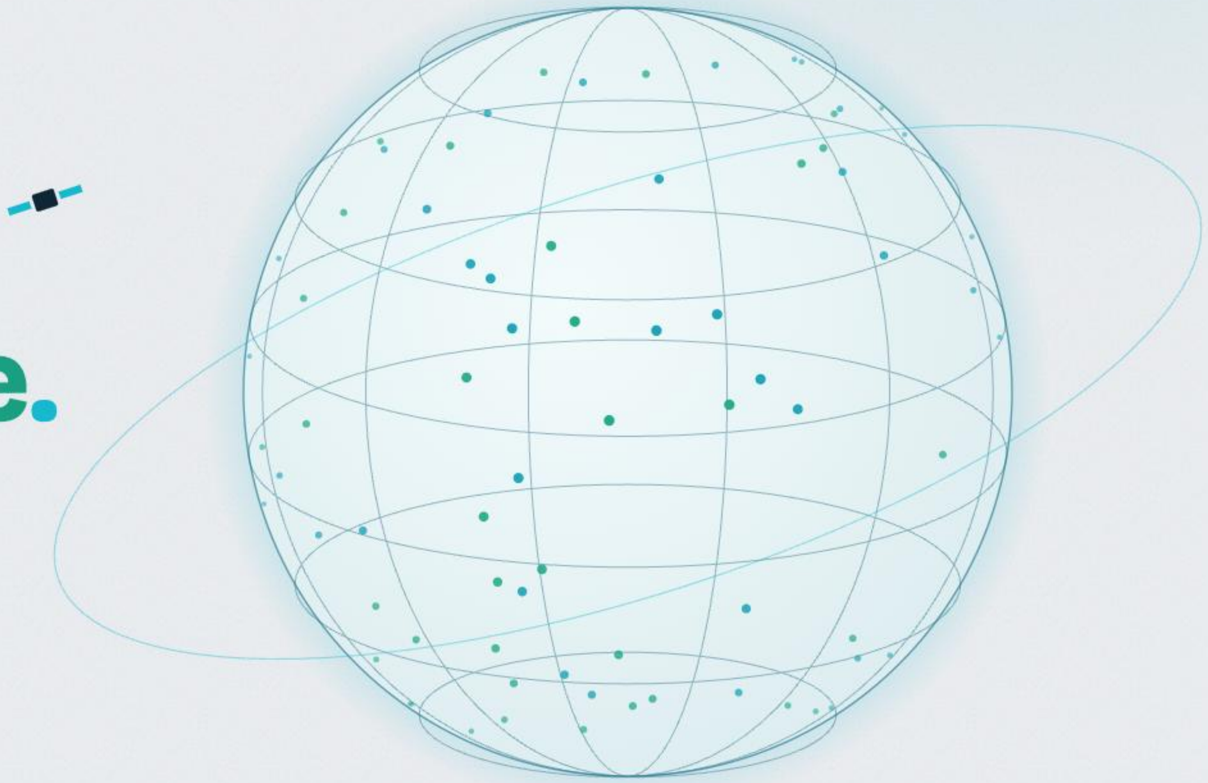
ACIBADEM MAAÜ / ML·DL INTERNSHIP / 2026

Predicting Village Wealth from Space.

A fairness & uncertainty audit of satellite poverty mapping — across 23 sub-Saharan African countries.



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23 COUNTRIES · 36,090 VILLAGES

What this is — and what it isn't

We reproduce Yeh et al. (2020) faithfully — then run the audit the original paper didn't.

✓ WHAT IT IS

- **A faithful replication of Yeh et al. (2020)**
  8-band satellite imagery → a ResNet-18 → village wealth, on the same data family.
- **The audit the original never ran**
  We then ask whether that model is **fair, calibrated, and robust** — the questions Yeh left open.
- **Honest, cross-country evaluation**
  Every number is measured on countries the model **never trained on**, across all 23.

✕ WHAT IT ISN'T

- **Not a new architecture or SOTA claim**
  The network is a standard ResNet-18 — the contribution is the **evaluation**, not the model.
- **Not a deploy-ready targeting tool**
  This is a scientific audit, not a system to hand an aid agency tomorrow.
- **Not a criticism of the original**
  We build on Yeh 2020 and **reproduce its headline** before stress-testing it.

THE THREE QUESTIONS THIS TALK ANSWERS

01 Fairness
Is it as accurate for the **rural poor** as for rich cities?

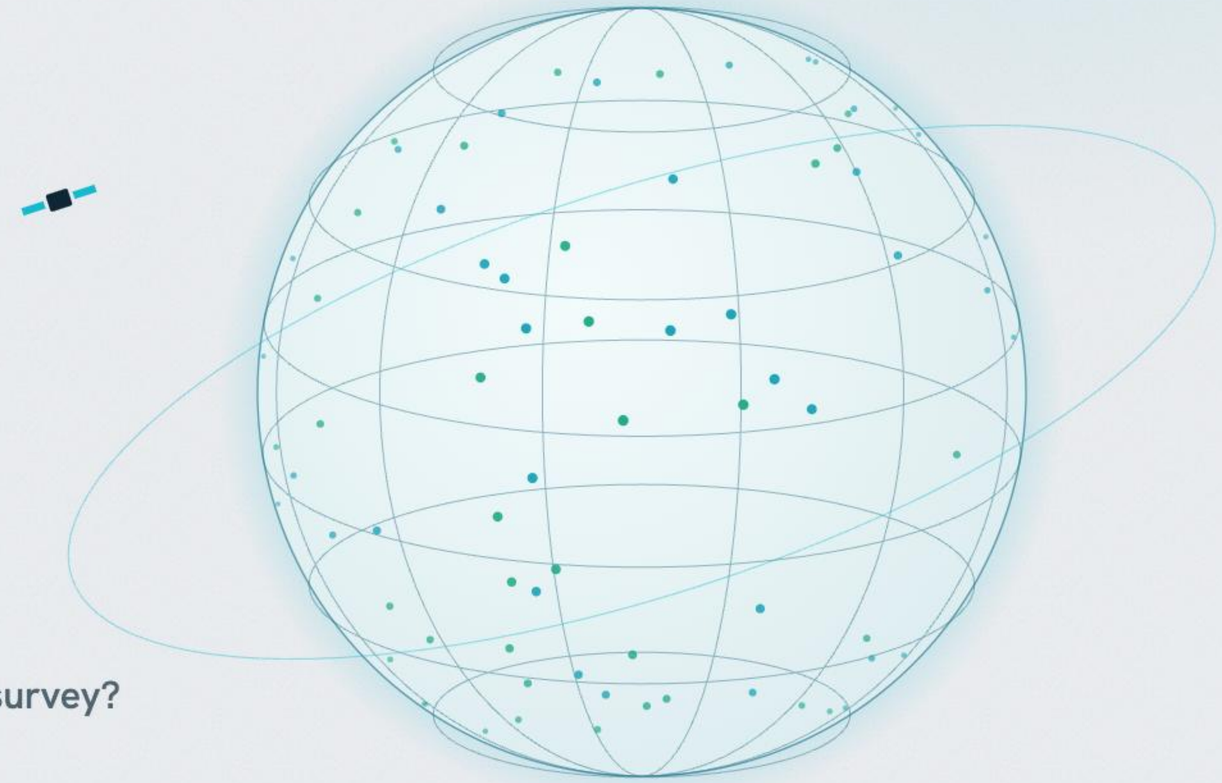
02 Uncertainty
When it's wrong, does it at least **know it's unsure**?

03 Generalization
Does it still work in countries it has **never seen**?

01

The Promise

Can a **free satellite image** replace an expensive household survey?



Finding the poor is slow, costly, and out of date

To help people in poverty, you first have to **find** them — and the ground-truth data to do that is scarce.

WHY IT'S HARD

Household surveys are the gold standard, but they're **expensive, slow, and sparse** — many regions are surveyed only once every several years, if at all.

THE OPPORTUNITY

Satellites image **the entire planet, for free, repeatedly**. If a model can read wealth from that imagery, it becomes a scalable **complement** to surveys — filling the gaps in time and space.

THE STAKES

These maps already steer **real aid**: Togo's Novissi program used satellite poverty maps to target emergency COVID cash (Aiken 2022). When the map is wrong about the poorest, real people are missed.

THE NEED

1.8B

people below the poverty line — and the data to **locate** them is scarce and stale

THE DATA GAP

~5 yrs

typical gap between household surveys — **often far longer** in the poorest places

The idea: one model, one number

The whole system in one line — then we spend the rest of the talk unpacking every piece honestly.

1

GPS

Every surveyed village comes with a GPS coordinate.

2

Satellite tile

We pull a 6.7 km image centred on that point.

3

8 bands

Colour · infrared · thermal · **night-lights** — eight ways to see it.

4

ResNet-18

A standard CNN reads all eight channels together.

5

One number

It outputs a single value: the predicted **wealth**.

WHY IT CAN WORK

Wealth has a **physical footprint** visible from above — dense metal roofs, paved roads, and electric light at night. The CNN learns to read those cues into a single score.

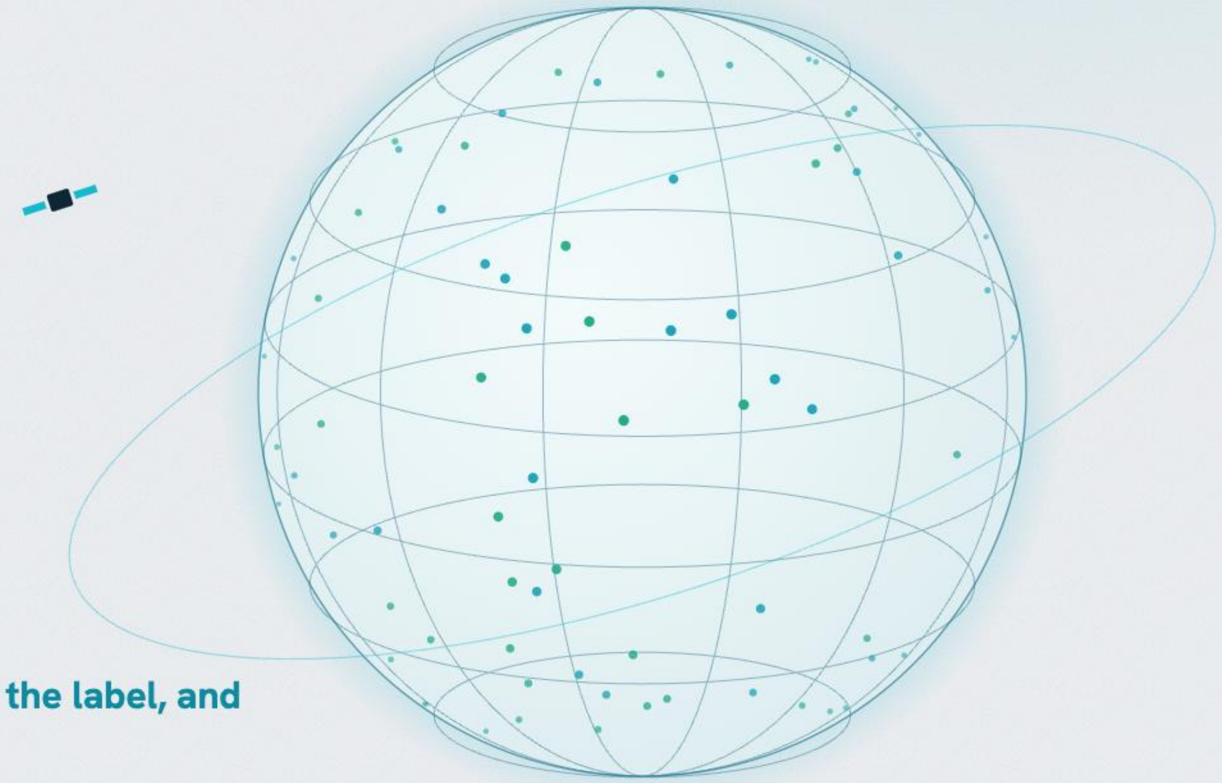
WHAT IT LEARNS FROM

Every prediction is **trained to match the DHS wealth index** — the “answer key” we build with PCA in the next section. No survey label, no training signal.

02

The Ingredients

Where the wealth number actually comes from — **the data, the label, and the image.**



The answer key: DHS household surveys

Our model is only as good as its ground truth — so where does the “true” wealth come from?

WHAT IT IS

The **Demographic & Health Surveys** are gold-standard, nationally-representative household surveys. A survey **cluster ≈ a village** (~25 households), and each cluster has a **GPS coordinate**.

THE DELIBERATE TWIST

For privacy, DHS **randomly shifts** each GPS — up to **2 km** in cities, **5 km** in rural areas. So the coordinate we get is deliberately **imprecise**.

HOW WE HANDLE IT

That’s exactly why our tile is a wide **6.7 km box** — big enough that the true village is still inside it, even after the shift. We trade resolution for **guaranteed coverage**.

A wide tile still contains the real village — despite the shift.

First we clean the data — then prove how clean it is

Every modelling result rests on the data underneath it, so the cleaning is explicit — and measurable.

MISSING VALUES

DHS “don’t know” codes (96–99) become NaN, then imputed by the **within-country mean** — a missing answer never reads as a real zero.

COVERAGE FILTER

We drop households missing more than **30%** of their asset fields — too little signal to place them reliably.

GPS QUALITY

Clusters at “null island” (0, 0) or with broken coordinates are **removed** before any imagery is matched.

THE IMAGERY

Any cloud or missing pixel in a tile is **mean-filled per channel** — the network never sees a hole.

HOW COMPLETE IS EACH SATELLITE TILE?

- **95% of tiles** — under 5% missing pixels (essentially complete)
- **5% of tiles** — 5–19% missing, mean-filled per channel

0.66% mean missing per tile	**95%** of tiles under 5% missing	**19%** worst single tile (still 81% intact)

→ So the gap-filling is negligible — the network trains on **genuine imagery, not invented pixels.**

There is no “wealth” column

DHS never measures wealth directly — so, like a **credit score**, we let PCA **discover** it from the asset data.

WHAT IT IS

Principal Component Analysis finds the single weighted mix of many correlated variables that captures the most variation. That mix is the first principal component — **PC1**.

WHAT IT'S FOR

It’s the standard way to collapse dozens of indicators into one meaningful score — turning a table of yes/no answers into a single comparable **index**.

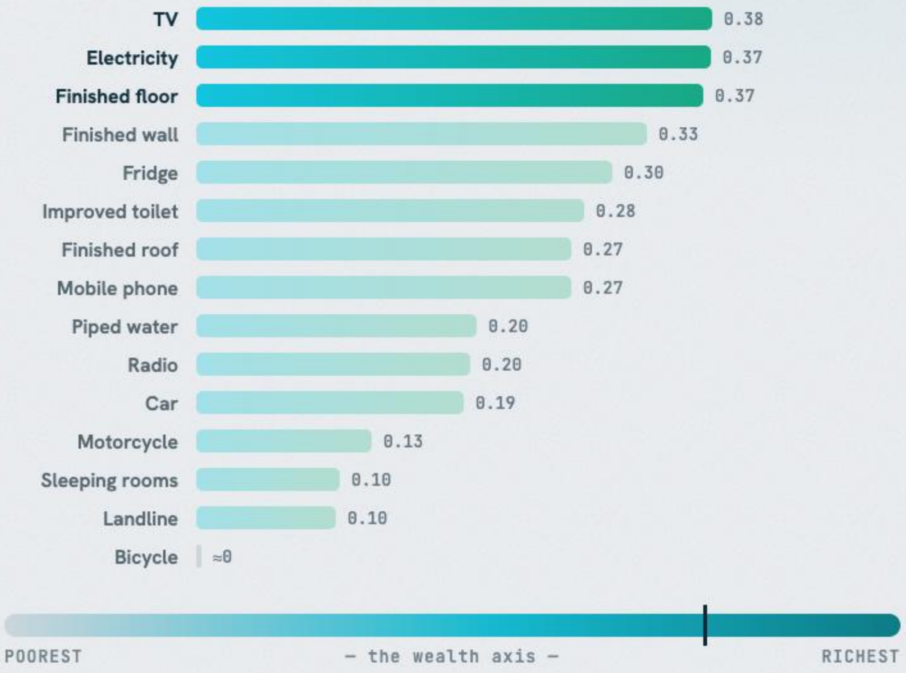
HOW WE USED IT

We ran PCA on ~15 DHS **asset variables** across all households. PC1 is the axis they differ on most — wealthy homes score high, poor homes low. Each household’s PC1 score is its wealth index; we **average it to the village** (≈25 homes).

WHAT IT MEANS

PC1 holds **28.7%** of all variation (3× the next) and **14 of 15** assets pull the same way → an unambiguous “more assets = wealthier” axis. Built with the standard DHS asset-index method (Rutstein & Johnson), **pooled across countries** for one common scale.

ASSET LOADINGS ON PC1 — THE “WEALTH WEIGHTS”



28.7%
of all variance in one axis

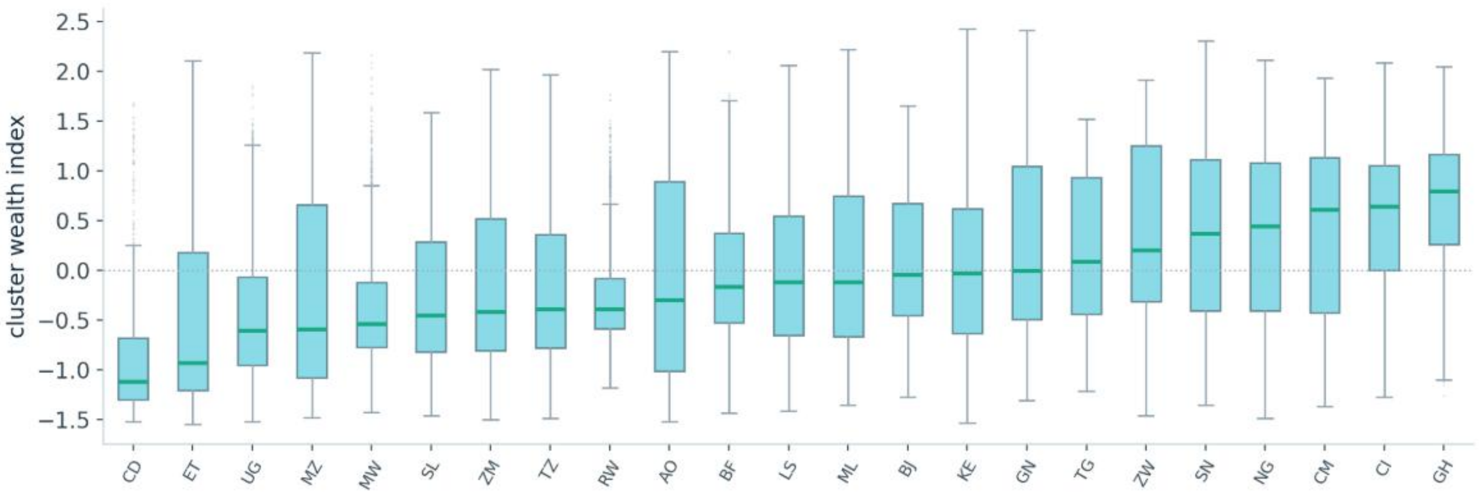
3×
larger than the next component

14 / 15
assets load positive (bike ≈ 0)

The data at a glance

Box plots show the **spread** of village wealth — and the one split that drives the second half of this talk.

WEALTH INDEX BY COUNTRY — POOREST (DRC) → RICHEST (GHANA)



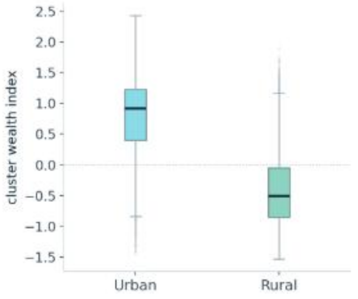
HOW TO READ IT

Each box = the middle 50% of villages; the line is the median; whiskers span the range; dots are outliers.

WHY IT MATTERS

Wealth varies **within** every country, with heavy overlap between them — so this is **not** a trivial 'guess the country' task.

URBAN vs RURAL



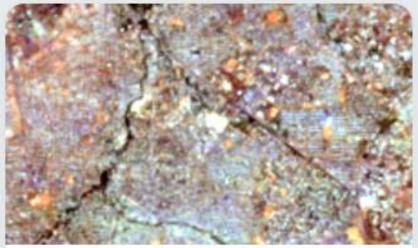
THE SPLIT THAT MATTERS

Urban villages sit far above rural ones — median **+0.91** vs **-0.51**. Rural wealth is bunched near the bottom. This urban-rural gap is the thread we pull on in the fairness section — the model will turn out to serve cities better than villages.

Eight channels — eight ways to see a village

Not eight photos — **one image, eight channels deep** (like RGB, but with 8). Each channel exposes a different physical signature of wealth.

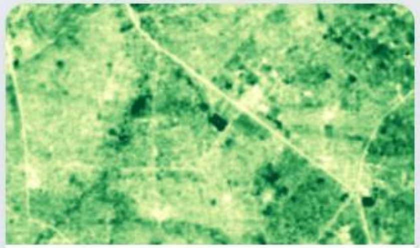
↓ ALL FIVE ARE THE SAME VILLAGE — ONE URBAN CLUSTER IN MALI, SEEN THROUGH ITS EIGHT BANDS



Colour

R · G · B

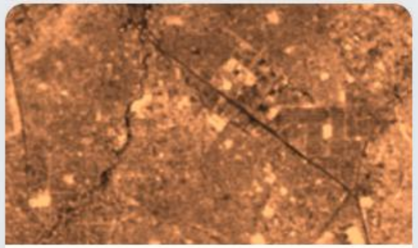
What your eye would see — roofs, roads, the layout of the settlement.



Near-infrared

NIR

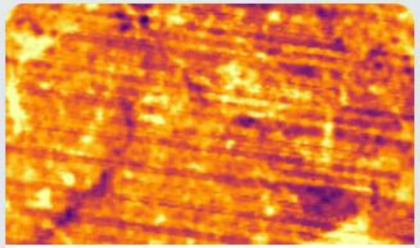
Vegetation vigour and rooftop materials the eye can't see.



Shortwave IR

SWIR 1·2

Built-up surfaces, bare soil, and moisture.



Thermal

TEMP

Surface heat — denser, built-up areas read hotter.

Night-lights

NL

Electrification at night — the **strongest** single wealth cue.

HOW EACH CHANNEL IS MADE

Each one is the **median over a 3-year window** — clouds average out into one clean image. Source: **7 Landsat bands (30 m) + VIIRS night-lights**.

WHY EIGHT, NOT THREE

Colour alone misses a lot. The CNN **fuses all eight** into one number — and night-lights will turn out to do most of the work, which matters later.

The model, in 20 seconds

Deliberately ordinary — because the contribution of this work is not the network.

WHAT IT IS

An off-the-shelf **ResNet-18** — the same network used for ordinary image classification.

WHAT WE CHANGED

Its first layer accepts **8 channels, not 3**; its last layer outputs **one number** (regression), not 1,000 class labels.

HOW WE TRAIN IT

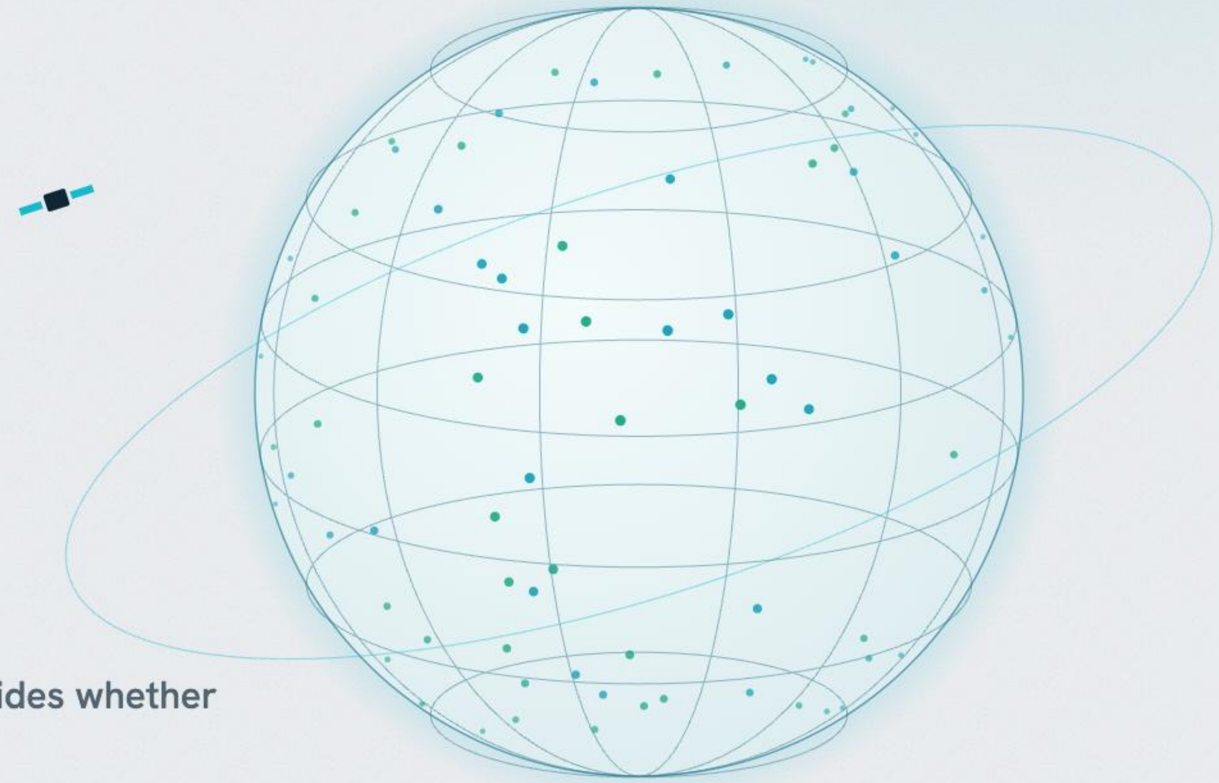
- **MSE loss** — optimises r^2 directly
- **Adam, lr 3e-4** (cosine) — stable, low-tuning; cut from 1e-3
- **Batch 64** — fits the 8-channel tiles in GPU memory
- **Early-stop on val r^2** — best-generalising model
- **From scratch** — pre-training doesn't help (backup)

The science is in the data and the honest evaluation — not the architecture.

03

Does it work?

First — **the honest test.** Because how you split the data decides whether the number is real.



Three numbers, three questions

Before any results — the three ways we grade the model, in plain English.

PEARSON r

Do predictions track the truth?

Plot predicted against true wealth: how tightly do the dots hug a **straight line**? A clean line scores high, a fuzzy cloud scores low — it's the strength of the linear trend.

`-1 → +1 · the metric the benchmarks report`

R^2

How much do we explain?

Of all the ways villages differ in wealth, what **share** does the model capture? **0** = no better than guessing the average for everyone; **1** = perfect. Hit hardest by big misses at the extremes.

`0 → 1 · 0.57 = we explain 57%`

SPEARMAN ρ

Did we get the order right?

Forget the exact numbers — line the villages up poorest→richest by truth, and by prediction. Do the two line-ups **agree**? Squeeze or stretch the values all you like; only the order counts.

`-1 → +1 · what targeting actually needs`

→ **Pearson and r^2 grade the values; Spearman grades the order — and for finding the poor, the order is what matters.**

Why a random split lies — and how we test honestly

The more honest the test, the lower the score. We climb the ladder and report the honest rung, not the flattering one.

RANDOM SPLIT

0.73 r^2

Shuffle all villages, then test on held-out ones — the model can **memorise each country**. Flattering, but fake.

× inflated — we don't use this

LEAVE-COUNTRY-OUT

0.57 r^2

↓ the honest number — a random split would inflate it to 0.73

Hold out **whole countries** — test on places the model never saw, with country-bootstrap 95% CIs. **Every number in this talk is measured this way.**

✓ what we report

OUT-OF-DISTRIBUTION

0.48 r^2

Brand-new countries — this **pooled** r^2 hides a per-country collapse on the rich tail (slide 20).

hardest test

EASIEST / MOST FLATTERING ------→ HARDEST / MOST HONEST

Analogy: a random split is an exam with questions you **studied**; leave-country-out is questions you've **never seen**. We grade ourselves on the second one.

The catch: same order, squeezed values

A model can rank villages **perfectly** yet get every number **wrong**. Watch what ‘regression to the mean’ does — it’s the seed of the whole second half.

SAME ORDER, SQUEEZED VALUES — FIVE NAIROBI AREAS

village	true wealth	model
Kibera	−1.5	**−0.9**
Mathare	−0.7	**−0.5**
Dagoretti	0.1	**0.0**
Karen	0.8	**0.6**
Runda	1.6	**1.0**

THE VALUES GET SQUEEZED

The model pulls every guess **toward the middle** — the poorest predicted less-poor (−1.5 → −0.9), the richest less-rich. The numbers are off, so **r^2 and Pearson suffer**.

THE ORDER SURVIVES

But poorest still ranks poorest, richest still richest — so **Spearman stays perfect**. That gap is exactly why we lead with **ranking**.

THE SEED OF THE BIAS

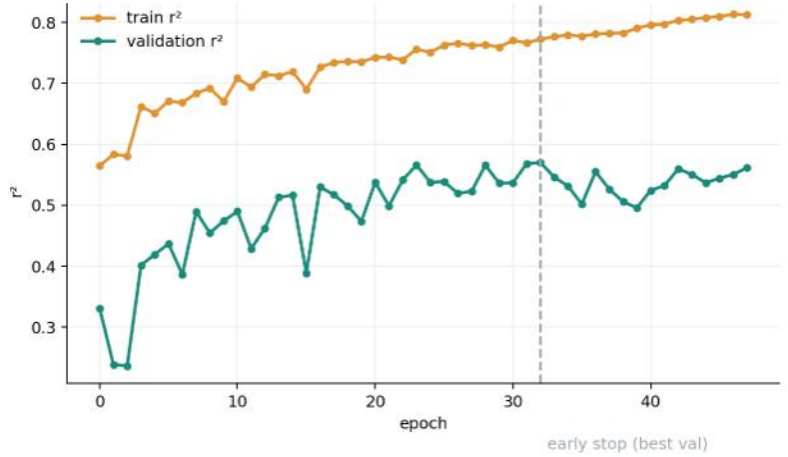
That squeeze toward the mean is **regression to the mean**. Harmless here — but on the real data it becomes the **staircase** that fails the poorest (slide 23).

→ **Perfect ranking, squeezed values — the first sign of the bias that defines the second half.**

Did it overfit? — the honest check

Two questions: does training run away from validation, and would more data help? Both answered from the curves.

LEARNING CURVE — TRAIN vs VALIDATION r² PER EPOCH



DATA-SCALING — VALIDATION r² vs TRAINING SIZE

WE STOP BEFORE IT DRIFTS

Training r^2 climbs to 0.81; validation plateaus near 0.55. We **early-stop at the best validation epoch** (dashed line) and only ever report validation / test — never the inflated training number.

THE LIMIT IS DATA, NOT THE MODEL

Give it more villages and validation r^2 keeps rising — **0.45 → 0.57**. The model is **data-limited**, not memorising noise. (More data lifts the *average*, though — not the poorest, slide 26.) A fancier net wouldn't help either: ImageNet-pretrained scores **–0.03 r^2**.

→ The train-validation gap is **controlled book-keeping** (early-stopping), and the ceiling is **set by data** — not overfitting.

Does it work? Yes — at benchmark level

On countries the model never trained on, it performs in the range of published satellite-poverty models.

0.76

PEARSON r

95% CI 0.72–0.79

Correlation with true wealth — **WILDS PovertyMap benchmark territory**, on a harder split.

0.57

R^2

95% CI 0.51–0.62

Share of wealth variation explained, **leave-country-out**.

0.72

SPEARMAN ρ

95% CI 0.67–0.76

Ranking — **the metric that matters** for targeting.

ON UNSEEN COUNTRIES

Every figure is **leave-country-out** with 95% CIs from a **country-level bootstrap** — the test countries were held out of training entirely.

HONEST, NOT INFLATED

A random split would have shown r^2 **0.73**. We report the country-blocked **0.57** — the inflation we refuse.

THE HALF THAT WORKS

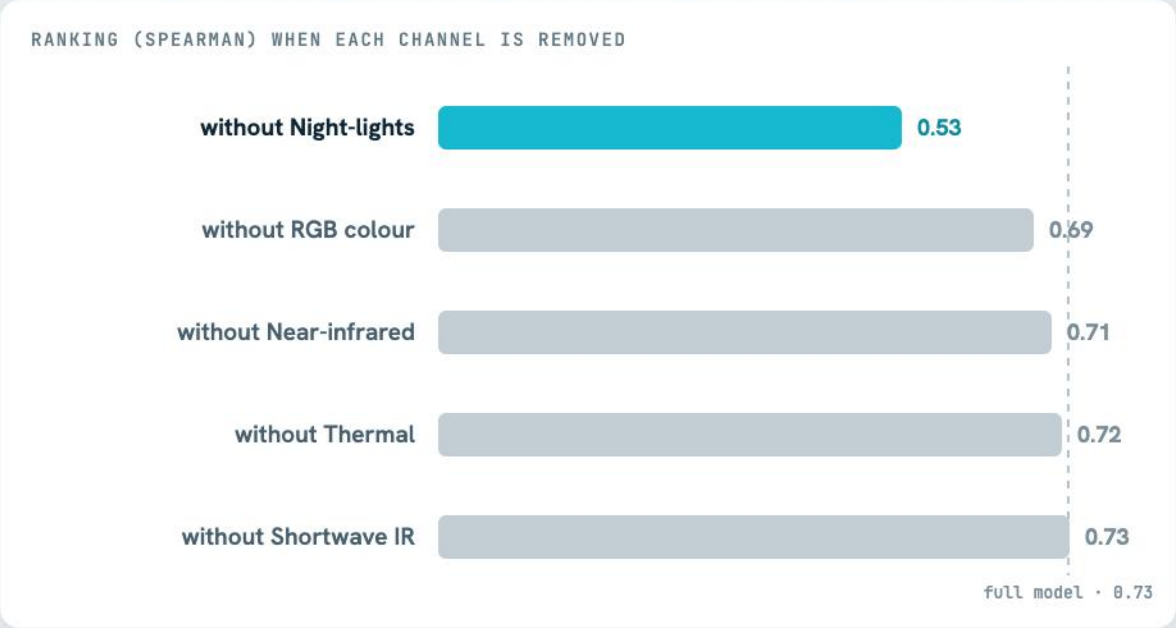
This is the 'it works' half of the story. The cracks — who it works *for* — come next.

HOW WE GOT THIS

Five-fold **leave-country-out** cross-validation: train on ~18 countries, test on the 5 held out, rotate until every country is tested once — then pool all 13,453 predictions. These are those pooled numbers.

What carries the signal? Night-lights.

Drop any single channel and ranking barely flinches — except one.



ONE CHANNEL DOMINATES

Remove colour, infrared, thermal or SWIR and Spearman stays ~0.70–0.73. Remove **night-lights** and it **collapses to 0.53** — far more than any other band.

WHY: ELECTRIFICATION

Lights at night are a direct proxy for **electricity access** — the strongest single signal of village wealth visible from space.

THE FORESHADOW

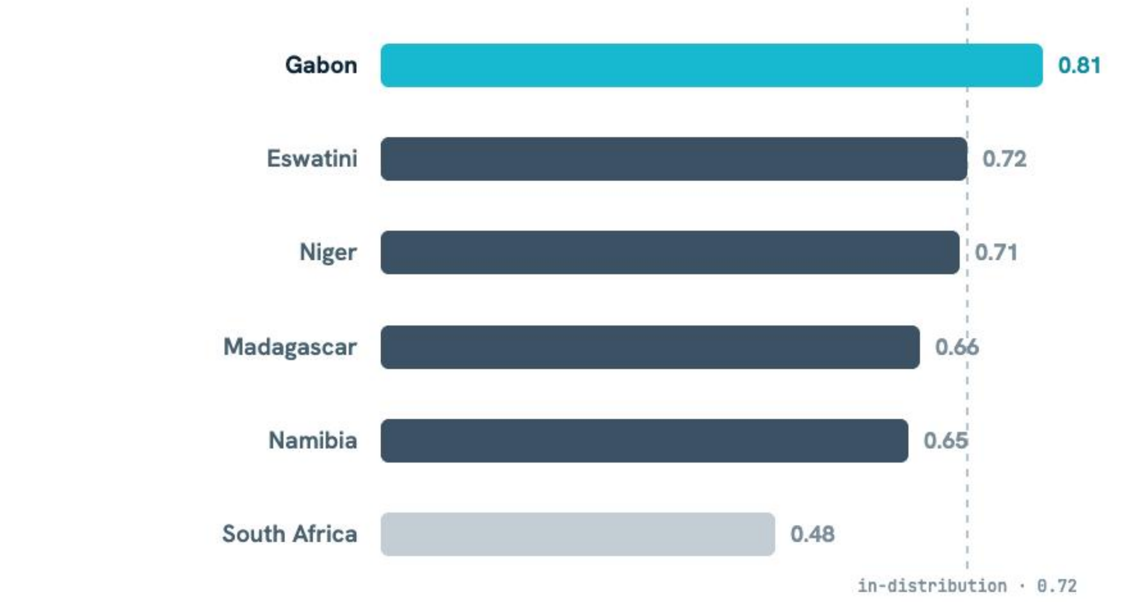
If the model leans this hard on lights — what happens in the places that have **none**? Hold that thought.

HOW WE GOT THIS

We **retrained the model once per channel group**, each time deleting that group from the 8-band input, and measured how far ranking fell. Only night-lights changed the answer.

It even generalises to countries it never saw

Frozen model, six brand-new countries outside the 23. Ranking transfers — but watch the extremes.



RANKING TRANSFERS

On countries entirely outside training, Spearman holds at 0.48–0.81. **Gabon (0.81) beats its in-distribution home (0.72)** — the ordering genuinely carries over.

BUT r^2 BREAKS

On the **wealthy extremes** the level falls apart: South Africa r^2 **−1.7**, Eswatini **−0.6**. It can rank villages but can't place where they sit.

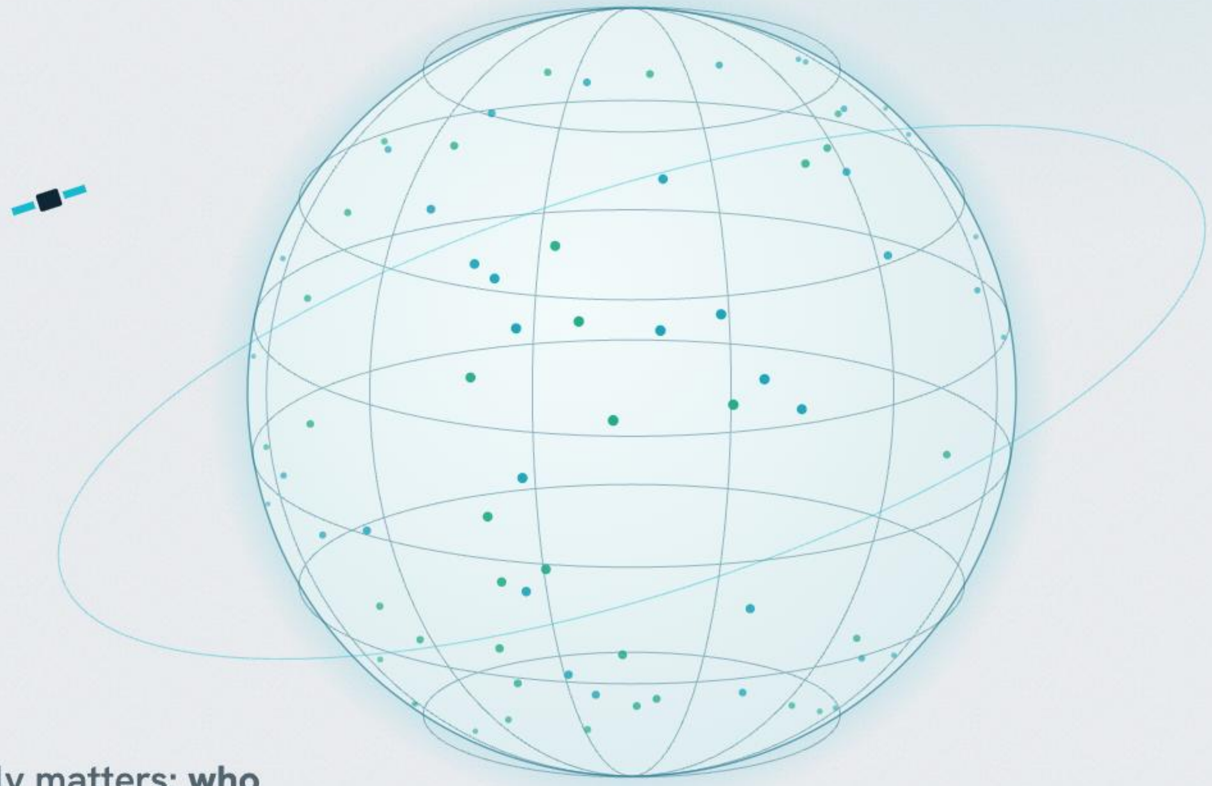
THE FIRST CRACK

Good for ranking, dangerous for absolute level — the exact split we dissect in the next act.

HOW WE GOT THIS We **froze the trained 23-country model** and ran it cold on six new countries — **no retraining, and not one survey point from them**. Zero data linkage, so this is genuine transfer.

04

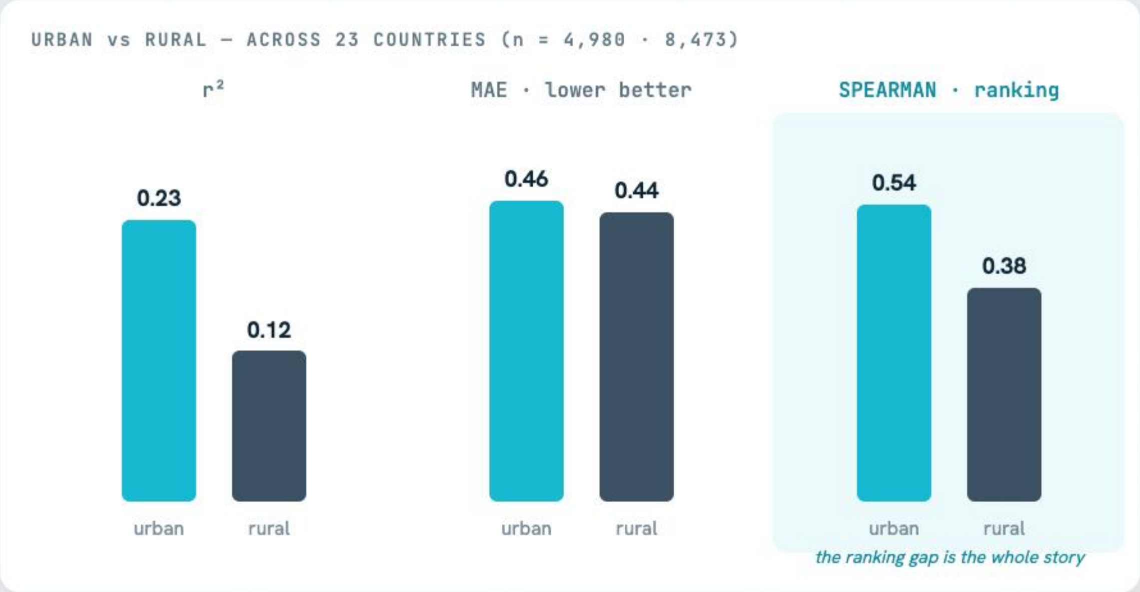
But it fails the people it’s meant to find.



It works — on average. Now we ask the question that actually matters: **who does it work for?**

It serves cities better than villages

It works on average — but averages hide who it fails. So we split every prediction into urban vs rural.



WHAT WE DID

Split all 13,453 held-out predictions by their DHS **urban/rural** tag and scored each metric within each group.

WHAT WE FOUND

Same *absolute* error (MAE 0.46 vs 0.44), but it can't **order** rural villages — Spearman **0.38 vs 0.54**, r^2 **0.12 vs 0.23**.

WHAT IT MEANS

Rural villages bunch near the bottom (slide 10), so errors stay small but the model can't resolve **order within** that band — which is why targeting later misses ~half (slide 29).

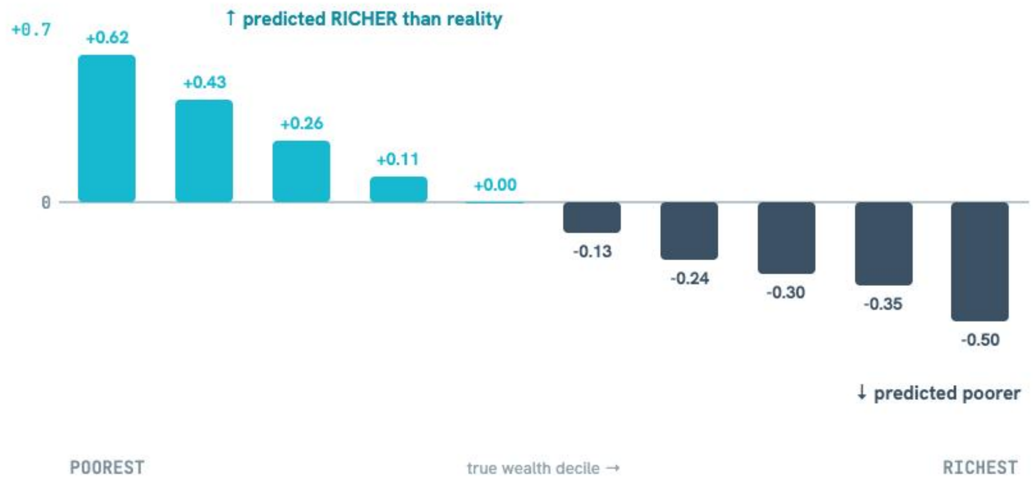
→ **Equal error, unequal ranking — it serves the rural poor worst.**

HOW WE GOT THIS Predictions tagged urban/rural by DHS cluster type; each metric computed within each group, pooled across seeds with bootstrap CIs.

Confidently wrong about the poorest

Does it fail the poor at random — or systematically? We measured its error at every wealth level.

PREDICTION BIAS (PRED − TRUE) BY TRUE-WEALTH DECILE



WHAT WE DID

Sorted all villages poorest→richest into 10 bins and averaged the **signed** error (predicted − true) in each.

WHAT WE FOUND

A clean staircase: it over-predicts the poorest by **+0.62** and under-predicts the richest by **−0.50** — slope 0.60.

WHAT IT MEANS

Classic **regression to the mean**: it drags every guess toward the average, lifting the poorest **out of the danger zone** on paper.

→ **It isn't noisy about the poor — it's reliably wrong, predicting them richer than they are.**

HOW WE GOT THIS Predicted regressed on true wealth across all 13,453 held-out villages (slope); signed error averaged within each true-wealth decile (bars).

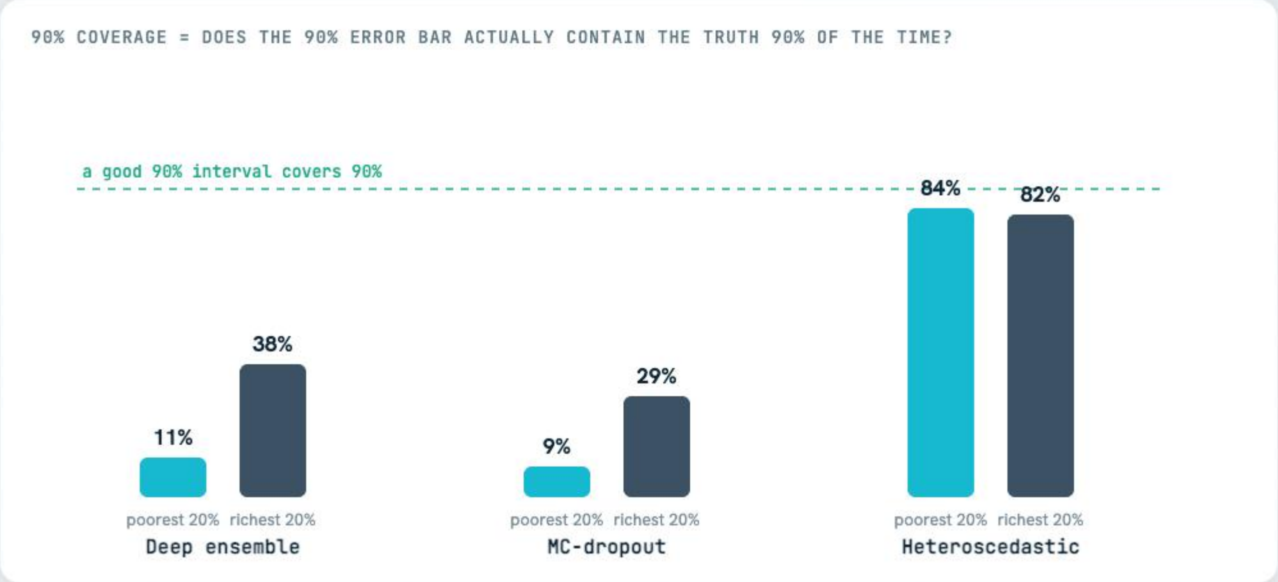
Does it even know when it's wrong?

Last hope: if it must fail the poor, can it at least flag its own doubt? We gave it three ways to express uncertainty, then checked its **90% error bars**.

DEEP ENSEMBLE
Train **5 copies** of the model; where they **disagree**, it's unsure.

MC-DROPOUT
Run **one** model many times with random neurons switched off; the **spread** is its doubt.

HETEROSCEDASTIC
The model predicts its **own error bar** for each village.



→ **Ensemble & MC-dropout cover just 9–12% of the poorest — overconfident. Heteroscedastic covers 84% vs 83% — calibrated, and equal. So — problem solved? →**

HOW WE GOT THIS Three uncertainty heads; 90% prediction-interval coverage measured separately for the poorest and richest 20%.

Calibration is not equity

The calibrated method looked like it fixed it. It didn't — and why is the most important result in this work.

THE CATCH — AURG ≈ 0

AURG (*Area Under the Risk-coverage Gain*) is how we test whether the model's uncertainty can **find its own mistakes** — do its 'most-unsure' guesses beat random at flagging the wrong ones? For all three methods it's ≈ 0. So even the 'calibrated' one hands the poor and the rich the **same** confidence — it can never single out the failures.

$$\text{error} = \text{noise} + \text{bias} + \text{variance}$$

UNCERTAINTY ONLY SEES VARIANCE

All three methods measure **variance** — how much predictions **wiggle** when you perturb the model. But for the poor, every model **agrees** on the same over-prediction, so variance is low and it looks **confident**.

THE POOR'S ERROR IS BIAS

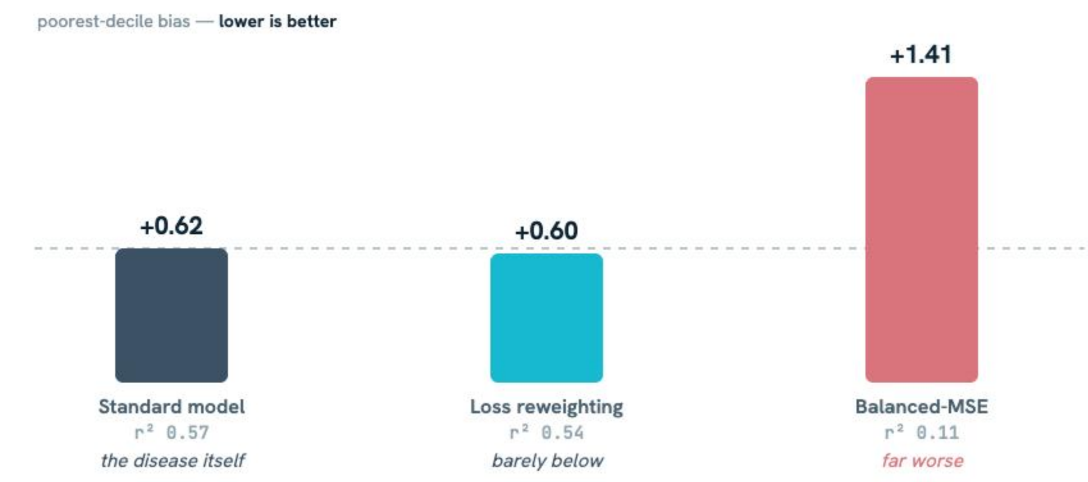
A **systematic, confident offset** — the staircase — not wiggle. Bias is **structurally invisible** to every uncertainty method, however well-calibrated.

→ **We hoped uncertainty could warn us about the poor. It can't — their error is bias, not variance. A model can be perfectly calibrated and still silently fail the poorest.**

We tried to fix it. Nothing works.

Is the poorest-bias a fixable tuning artifact — or something deeper? We ran the field's standard cures to find out.

POOREST-DECILE BIAS — STANDARD MODEL vs EACH ATTEMPTED FIX



WHAT WE TRIED

Starting from our standard model (the **+0.62-biased baseline**), we retrained it with the two go-to de-biasing losses — **reweighting** (LDS) and **Balanced-MSE** — and separately tripled the data.

WHAT HAPPENED

Reweighting nudged the bias **+0.62 → +0.60** as r^2 fell. Balanced-MSE wrecked accuracy (0.57→0.11) and made it **worse** (+1.41). More data lifted overall r^2 but left the poorest untouched.

WHAT IT MEANS

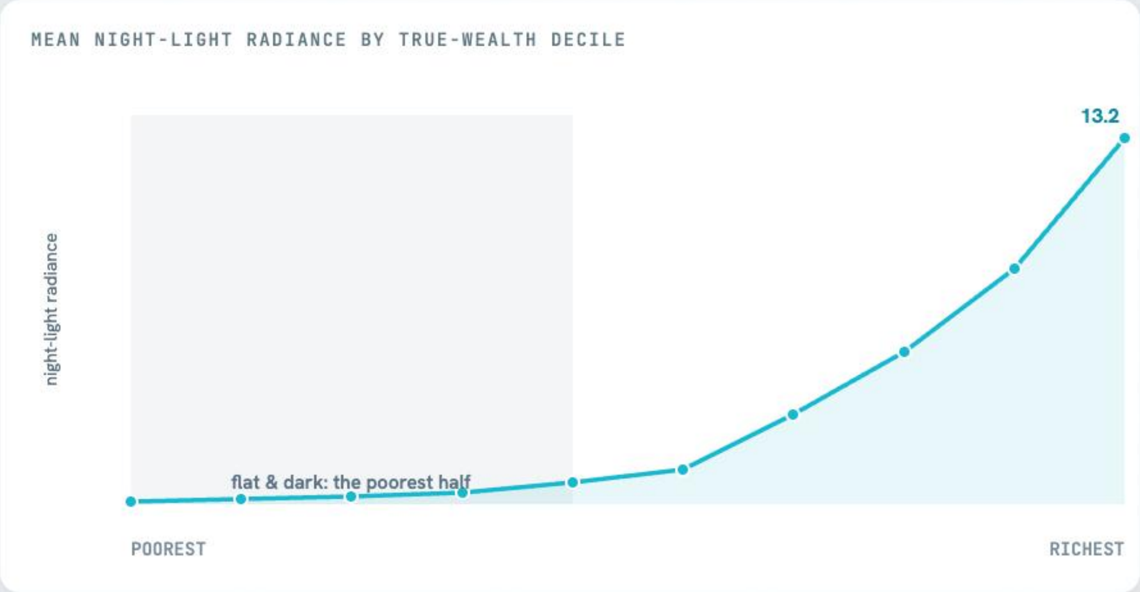
A slope is trivially re-scaled, but no loss can **invent ranking signal the imagery lacks** among the poor. The limit is the data, not the recipe — as the next slide shows.

→ The level is correctable, but no fix recovers the ability to rank the poorest — a signal limit, not a tuning problem.

HOW WE GOT THIS We retrained the baseline with LDS reweighting and a Balanced-MSE objective, then re-measured the same poorest-decile bias. Tripling the data doesn't move it either.

Why? Because the lights are dark

Reweighting and uncertainty can't fix it — so **why** does it fail the poor at all? We looked at its strongest signal.



WHAT WE DID

Averaged each village's **night-light brightness** by wealth decile, and measured the night-light-wealth correlation overall vs. within the poorest 30%.

WHAT WE FOUND

Flat and near-zero across the poorest half, then it rockets up. Lights track wealth **0.76 overall** — but only **0.28** among the poorest.

WHAT IT MEANS

The model's dominant signal is **physically absent** for the poor — it has nothing to read, so it hedges to the mean.

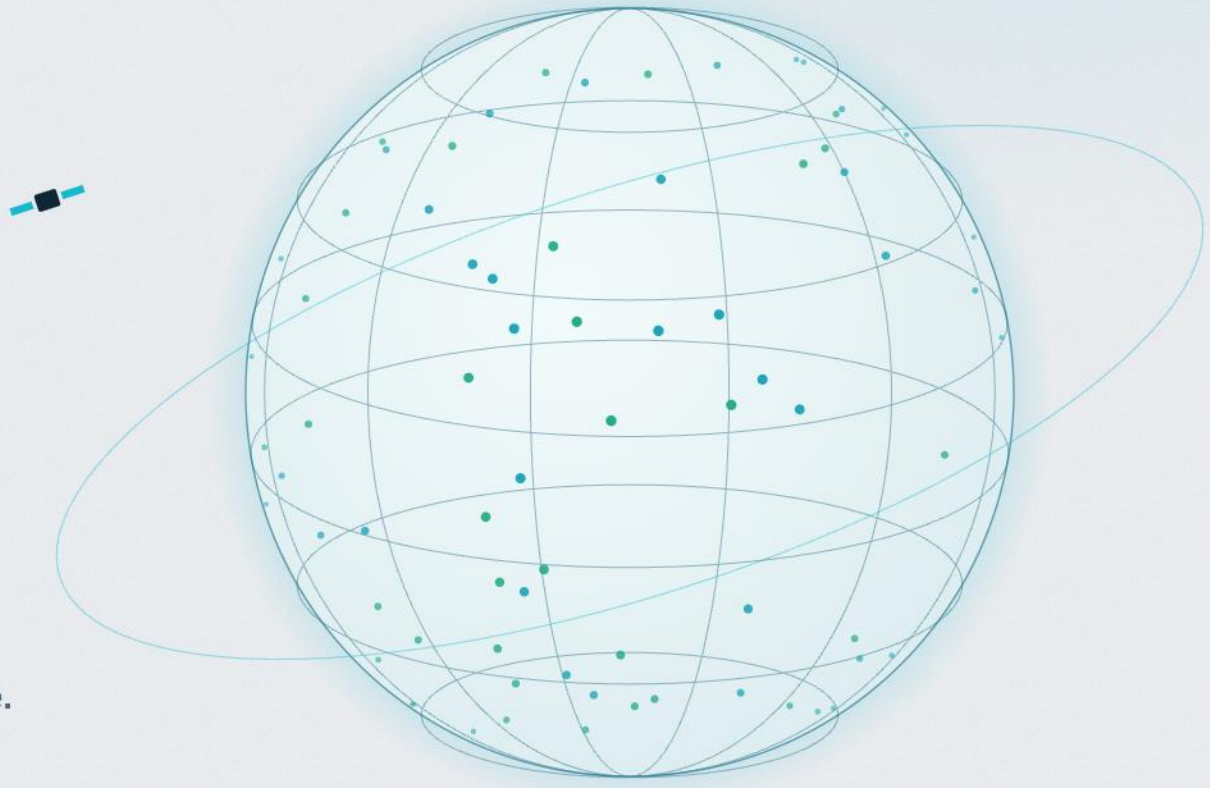
→ **The main signal is dark for the poor — so the ranking failure is fundamental: no loss can fill a gap in the data.**

HOW WE GOT THIS Night-light band averaged over the 3-year composite; villages grouped into true-wealth deciles; rank correlation measured overall and within the poorest 30%.

05

So what?

What it means for actually **using** these maps to reach people.



If you used it to target aid...

These maps already steer real money. So we asked the only question that matters in practice: who would you actually reach?

OF THE TRULY-POOREST 20% OF VILLAGES — WHO TARGETING BY THE MODEL REACHES

Tighter still: target only the poorest **10%** and you reach just **39%** — you miss **61%** of them.

WHAT WE DID

Ranked every village by **predicted** wealth, took the poorest 20% (who'd get aid), and checked how many of the **truly**-poorest 20% are in it.

WHAT WE FOUND

Just **49%** — 2.5× better than random (20%), but still half the neediest are ranked out of the aid bracket.

WHAT IT MEANS

Aid would skew toward the **less-poor**, and — from the urban/rural split — the **rural poor** are the ones missed most.

→ **Target the poorest with this map and you'd miss about half of them — the rural poor most of all.**

HOW WE GOT THIS Poorest-20% targeting recall computed on the 13,453 held-out villages: overlap between the model's poorest quintile and the true poorest quintile.

THE ONE THING TO REMEMBER

Satellite poverty maps are excellent for ranking regions — and dangerous for targeting the poorest.

✓ Trust it to

Compare whole regions — **which districts are poorer than others**. Ranking transfers, even to new countries.

✕ Don't trust it to

Find the neediest villages — it **misses half**, is biased against the poorest, and **can't tell you when it's wrong**.

What’s new here

An audit, not an architecture — three layers, from solid foundation to the genuinely novel.

THE FOUNDATION

Faithful replication
Yeh 2020 reproduced at benchmark level on a harder, country-blocked split.

Night-light dominance
Confirmed the single signal carrying most of the prediction.

THE AUDIT (extends prior work)

23-country fairness
Urban>rural ranking gap, scaled from Aiken’s 10 countries to 23.

Decile-resolved bias
The regression-to-mean failure mapped wealth-level by wealth-level.

Temporal + OOD validity
Holds out future rounds and six entirely unseen countries.

MOST NOVEL

Calibration ≠ equity
No uncertainty method flags the poor — because their error is bias, not variance. An equity-framed UQ negative result.

The root cause, measured
The dominant signal is physically dark for the poor — shown, not argued.

Honest framing: this is a rigorous audit and a negative result — no new state-of-the-art is claimed.

Thank you.

Questions — and a few numbers **in my back pocket:**

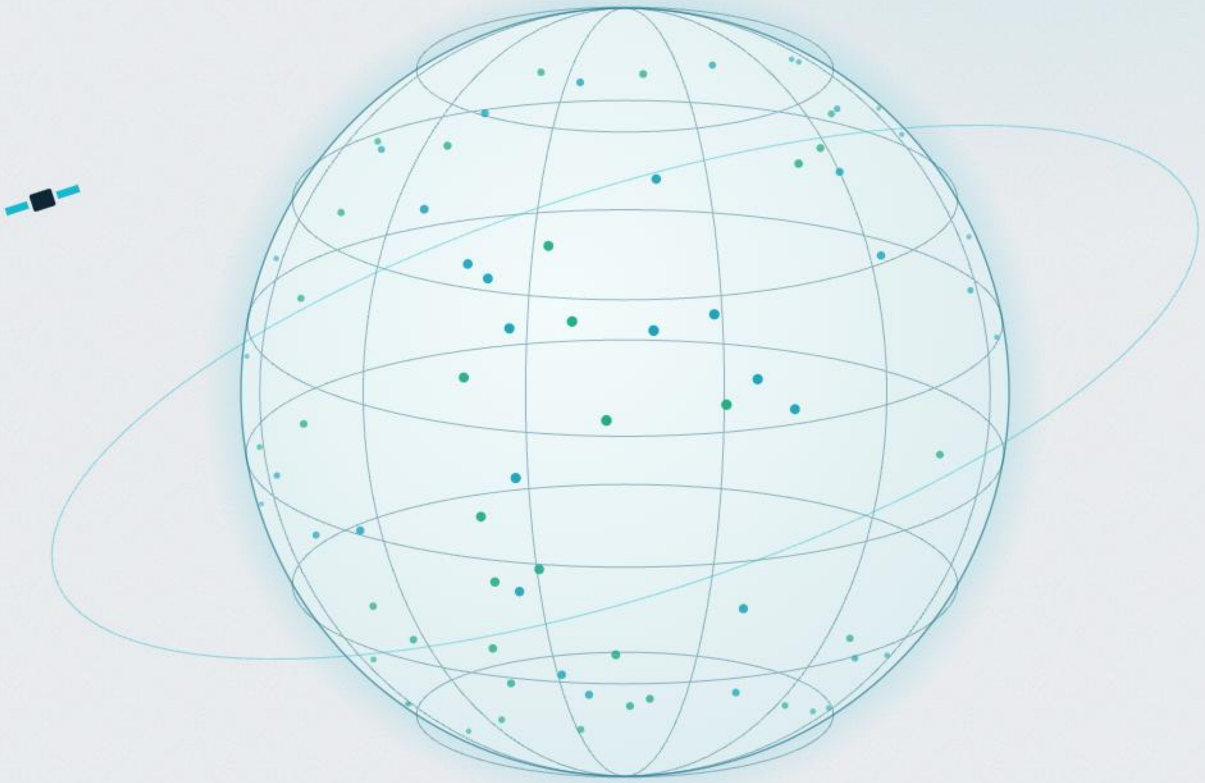
0.76
benchmark Pearson r

0.57
r^2 leave-country-out

+0.62
poorest over-predicted

0.28
NL-wealth, poorest 30%

49%
of the neediest reached



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Controls & robustness

The checks behind the headline claims — ready for the questions.

"DOESN'T A SIMPLE REGRESSION DO THIS?"

Night-lights-only: Spearman 0.76, r^2 0.16

A 1-feature night-light model **ranks** villages as well as the CNN — but a linear fit captures almost none of the **level** (r^2 0.16 vs 0.57). The CNN's value is the non-linear level, not the ranking.

"WOULD A FANCIER NETWORK HELP?"

Pretrained: −0.03 r^2

ImageNet pre-training makes it slightly **worse** than from-scratch. The gap to Yeh isn't the architecture — **it's the data budget**. A bigger network won't close it.

"DO YOUR CV FOLDS LEAK SPATIALLY?"

Residual Moran's I 0.18–0.47

Residuals are spatially autocorrelated **within** countries (expected), but we hold out **whole countries** — so cross-fold leakage is limited to border regions, not the bulk of the test set.

"ISN'T A SLOPE OF 0.60 JUST CORRECTABLE?"

Yes — the level, not the ranking

An affine rescale removes the +0.62 bias on paper. But it cannot recover **order among the poorest** — because the signal isn't there (slide 27). That's why we call it a signal limit, not a tuning bug.

Robustness: it survives within-country

The wealth index is pooled across 23 countries — so are the findings just the model sorting rich countries from poor? We removed the between-country signal and re-checked every claim.

RANKING HOLDS

0.70

within-country Spearman (vs 0.72 pooled)

It ranks villages **inside** a country almost as well as overall — it isn't just sorting countries.

SHRINKAGE IS REAL

0.65

within-country slope · 23/23 countries

Regression to the mean (slope < 1) shows up **within every country** — not a between-country effect.

POOREST-BIAS IS REAL

+0.41

poorest-20% bias · 21/23 countries

It over-predicts the poorest villages **inside each country** too — the failure is genuinely local.

→ **All three findings survive removing the between-country signal — they're real within-country, not artifacts of the pooled index.**

HOW WE GOT THIS Each claim re-evaluated against a within-country-standardised wealth index (between-country level removed), per country, then n-weighted across all 23 countries.